Module 10 Lab - Unsupervised Learning**Objective:** To understand and apply two common **unsupervised learning** techniques: **K-Means Clustering** for finding groups in data and **Principal Component Analysis (PCA)** for dimensionality reduction.**In this lab, you will write the code to perform clustering and PCA, and you will interpret the results.**

Part 1: Supervised vs. Unsupervised LearningSo far, every model we have built has been a **supervised learning** model. This means we had **labeled data**—we had both the features (X) and the correct answer, or target (y). The goal was to learn a mapping from X to y.In **unsupervised learning**, we only have the features (X). There is **no target variable**. The goal is to discover interesting structures or patterns hidden in the data on its own.Two major types of unsupervised learning are:* **Clustering:** Automatically grouping similar data points together.* **Dimensionality Reduction:** Compressing the data by reducing the number of features while trying to preserve as much information as possible.

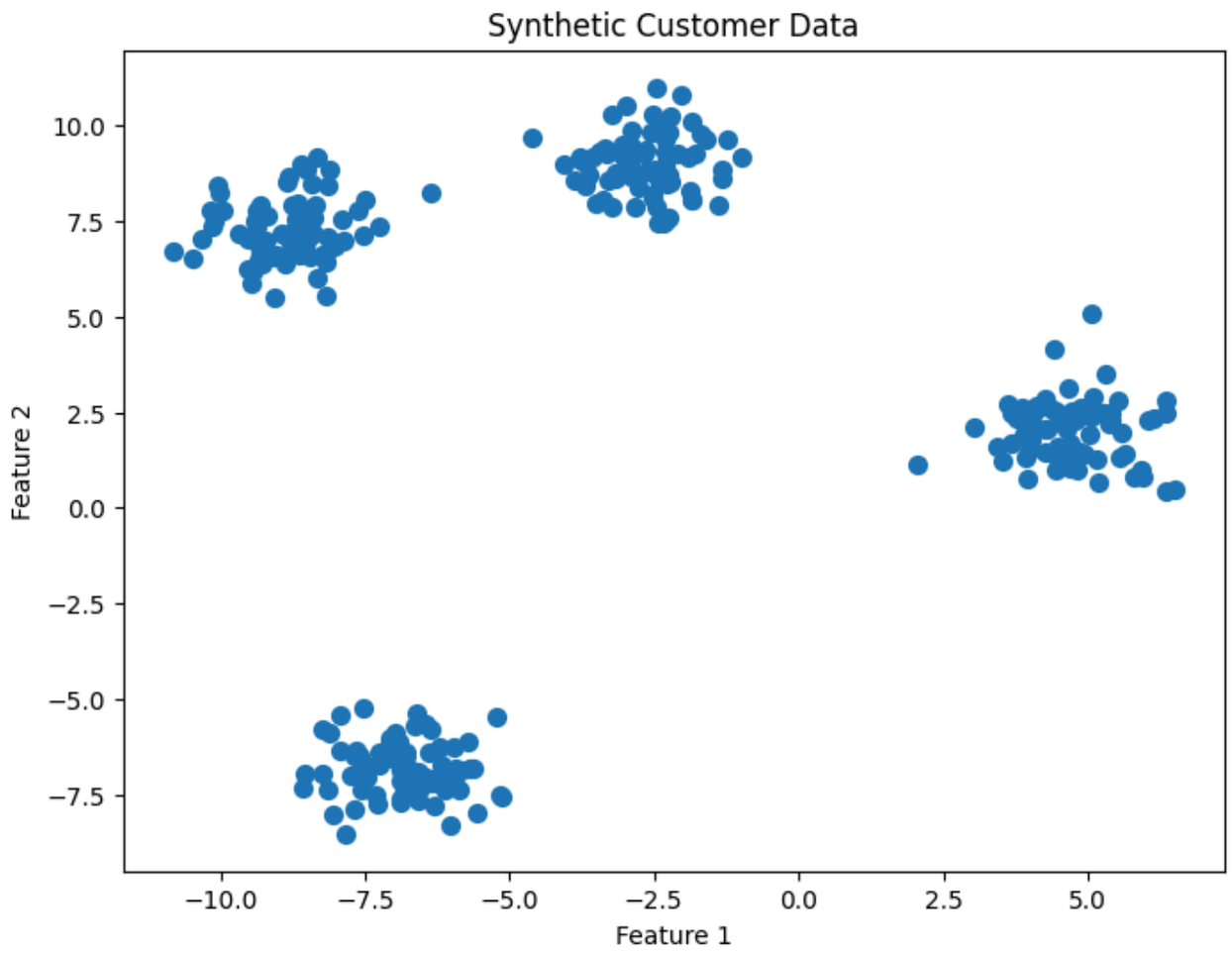
Part 2: K-Means Clustering**Concept:** K-Means is an algorithm that finds a pre-defined number of clusters (`k`) in a dataset. It works by:1. Randomly placing `k` centroids (the center of a cluster) in the data.2. **Assignment Step:** Assigning each data point to its nearest centroid.3. **Update Step:** Moving each centroid to the average position of all the points assigned to it.4. Repeating steps 2 and 3 until the centroids stop moving.**Problem:** We will use a synthetic dataset of customer spending habits to segment them into groups.

```
import matplotlib.pyplot as plt
from sklearn.datasets import make_blobs

# Generate synthetic data
X, y = make_blobs(n_samples=300, centers=4, cluster_std=0.8, random_state=42)

# Visualize the data
plt.figure(figsize=(8, 6))
```

```
plt.scatter(X[:, 0], X[:, 1], s=50)
plt.title("Synthetic Customer Data")
plt.xlabel("Feature 1")
plt.ylabel("Feature 2")
plt.show()
```



Task 1: Find the Optimal Number of Clusters (The Elbow Method)**Concept:** How do we know what `k` should be? The **Elbow Method** helps us decide. We run K-Means for a range of `k` values and for each `k`, we calculate the **inertia** (the sum of squared distances of samples to their closest cluster center). We then plot the inertia for each `k`. The "elbow" of the curve—the point where the inertia starts to decrease much more slowly—is a good estimate for the optimal `k`.**Your Task:** Calculate and plot the inertia for `k` values from 1 to 10.

```
from sklearn.cluster import KMeans

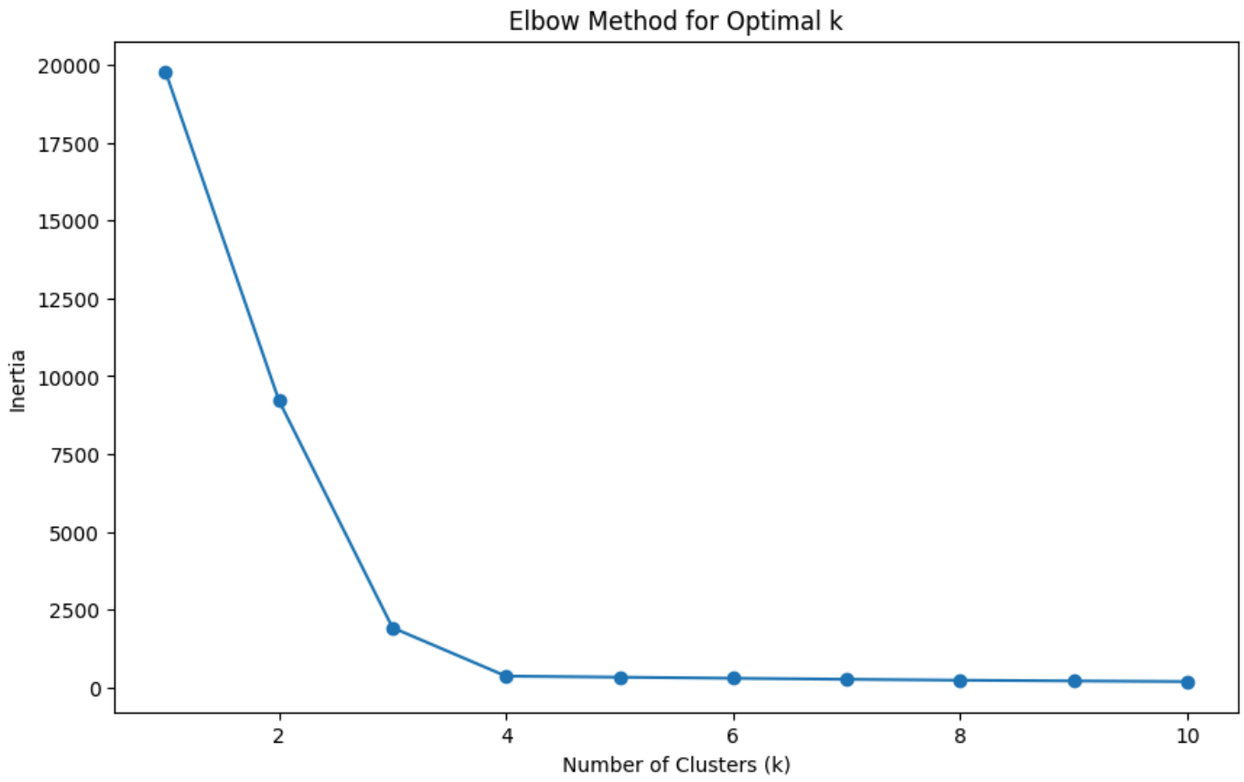
# 1. Create an empty list to store the inertia values
inertia = []

# 2. Loop through k values from 1 to 10
for k in range(1, 11):
    # 3. Create and fit a KMeans model with `n_clusters=k`
    kmeans = KMeans(n_clusters=k, random_state=42, n_init=10)
    kmeans.fit(X)

    # 4. Append the model's inertia_ to your list
    inertia.append(kmeans.inertia_)

# 5. Plot the results
plt.figure(figsize=(10, 6))
plt.plot(range(1, 11), inertia, marker='o')
plt.title('Elbow Method for Optimal k')
```

```
plt.xlabel('Number of Clusters (k)')
plt.ylabel('Inertia')
plt.show()
```



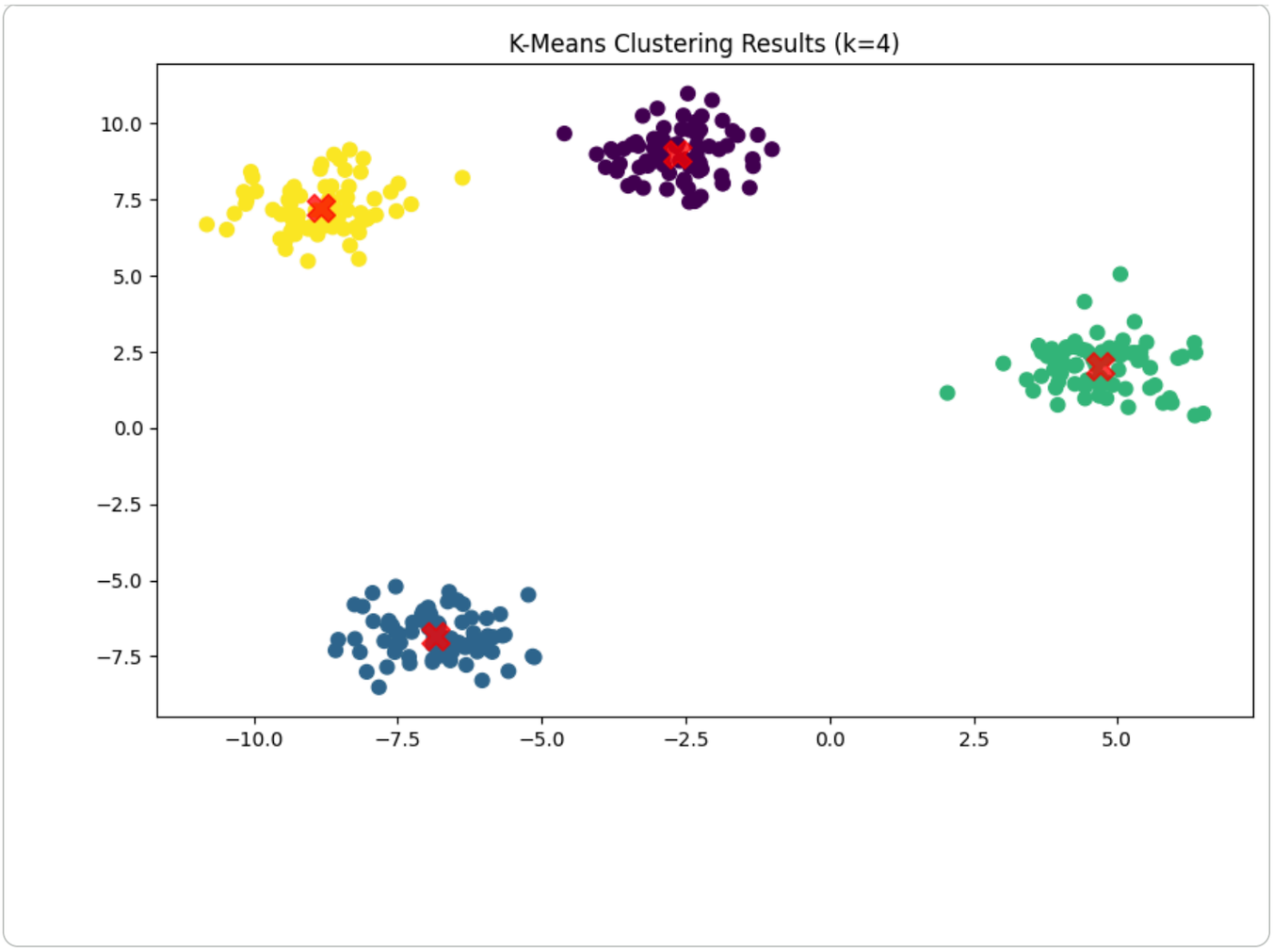
Task 2: Perform K-Means Clustering and VisualizeFrom the elbow plot, it looks like `k=4` is the optimal number of clusters.**Your Task:** Run K-Means with `n_clusters=4` and create a scatter plot showing the data points colored by their assigned cluster.

```
# 1. Create and fit a KMeans model with n_clusters=4
kmeans = KMeans(n_clusters=4, random_state=42, n_init=10)
kmeans.fit(X)

# 2. Get the cluster assignments for each data point
labels = kmeans.labels_

# 3. Get the coordinates of the cluster centers
centers = kmeans.cluster_centers_

# 4. Create a scatter plot of the data, colored by the labels
plt.figure(figsize=(10, 6))
plt.scatter(X[:, 0], X[:, 1], c=labels, s=50, cmap='viridis')
plt.scatter(centers[:, 0], centers[:, 1], c='red', s=200, alpha=0.75, marker='X')
plt.title('K-Means Clustering Results (k=4)')
plt.show()
```



Part 3: Principal Component Analysis (PCA)**Concept:** PCA is a dimensionality reduction technique. It transforms the data into a new coordinate system of **principal components**. These components are orthogonal (uncorrelated) and are ordered by the amount of variance in the data they explain. By keeping only the first few principal components, we can reduce the number of dimensions in our data while losing as little information as possible.**Use Cases:*** **Visualization:** Reducing high-dimensional data to 2 or 3 dimensions so we can plot it.* **Performance:** Speeding up model training by using fewer features.**Problem:** We will use the Iris dataset, which has 4 features, and reduce it to 2 features so we can visualize it.

Task 3: Apply PCA to the Iris Dataset**Your Task:** Use `PCA` from `sklearn.decomposition` to reduce the 4-dimensional Iris dataset to 2 principal components and visualize the result.

```
from sklearn.decomposition import PCA
from sklearn.datasets import load_iris
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# Load the data
iris = load_iris()
```

```
X_iris = iris.data
y_iris = iris.target

# 1. Create a PCA instance to reduce to 2 components
pca = PCA(n_components=2)

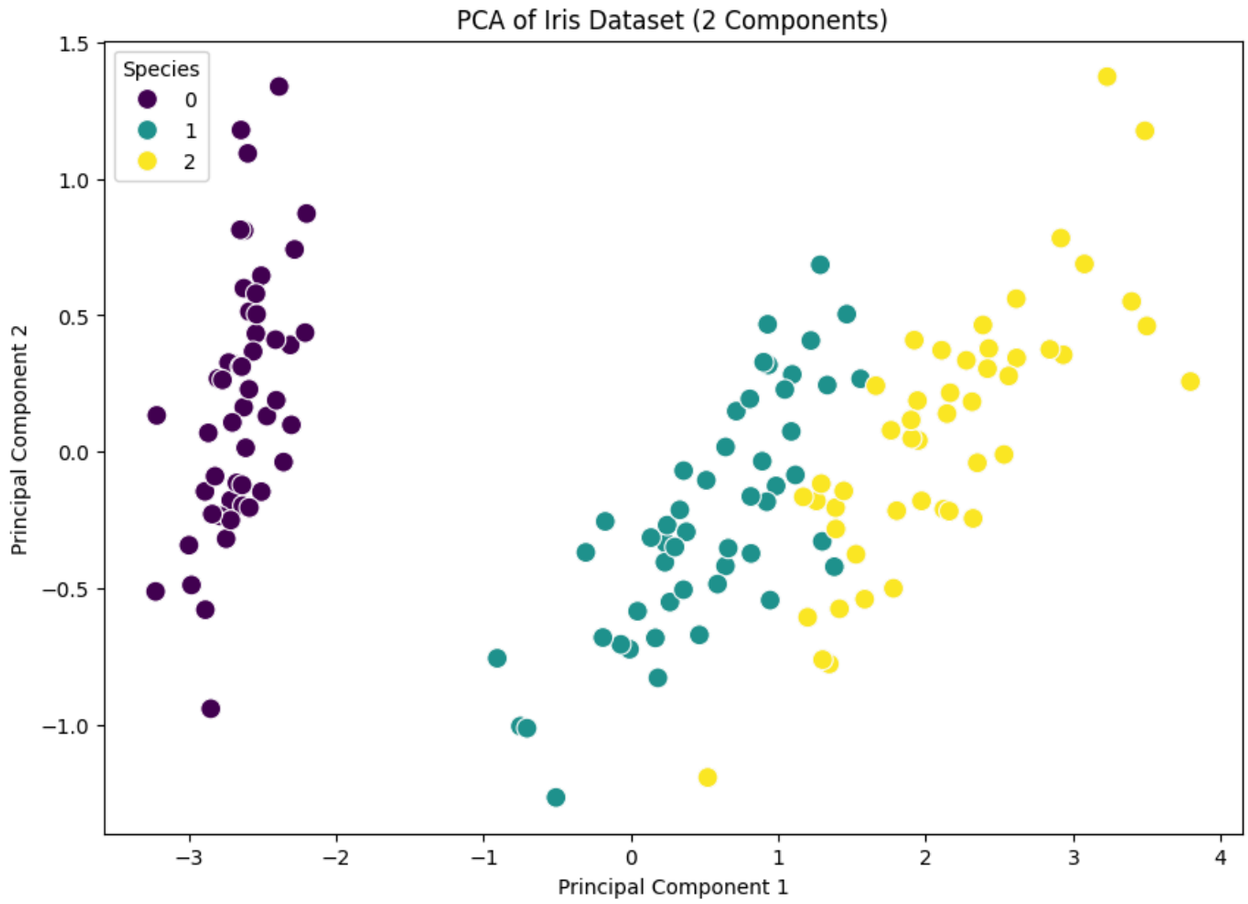
# 2. Fit PCA to the data and transform it
X_pca = pca.fit_transform(X_iris)

# 3. Print the explained variance ratio
print(f"Explained variance by component: {pca.explained_variance_ratio_}")
print(f"Total variance explained by 2 components: {pca.explained_variance_ratio_.

# 4. Create a DataFrame with the PCA results for plotting
pca_df = pd.DataFrame(data=X_pca, columns=["Principal Component 1", "Principal Cc
pca_df["Species"] = y_iris

# 5. Plot the results
plt.figure(figsize=(10, 7))
sns.scatterplot(x="Principal Component 1", y="Principal Component 2", hue="Specie
plt.title("PCA of Iris Dataset (2 Components)")
plt.show()
```

```
Explained variance by component: [0.92461872 0.05306648]
Total variance explained by 2 components: 97.77%
```



Close

```
## 📝 Knowledge Check
**Instructions:** Answer the following ques
**What is the fundamental difference betwee
learning?**
The process of supervised learning involves
made up of features vector X and the target
model in order to predict the outcomes accu
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📝 Knowledge Check

Instructions: Answer the following questions in this markdown cell.1.

What is the fundamental difference between

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unsupervised learning involves only the use

**2.In the K-Means elbow plot, why don't we
`k` to get the lowest inertia?**
The increase in k results in a decrease in
closer to the centroid of its respective cl
equal to the number of total data). The sel
in overfitting, as each datum will form its

**3.In the PCA results, the first two princ
of the variance.What does this tell you abo
dataset?**
From the findings, it is seen that most of
variations found within all four variables
dimensions. This shows the high inter-corre
which makes it possible to reduce the dimen
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supervised and unsupervised learning? The process of supervised learning involves the use of labeled data, which is made up of features vector X and the target variable y. This helps to train a model in order to predict the outcomes accurately. On the other hand, unsupervised learning involves only the use of unlabeled feature vector X.

2.In the K-Means elbow plot, why don't we just choose the largest possible `k` to get the lowest inertia? The increase in k results in a decrease in inertia, because every datum is closer to the centroid of its respective cluster (as it tends to zero when k is equal to the number of total data). The selection of the highest k will result in overfitting, as each datum will form its own cluster.

3.In the PCA results, the first two principal components explained over 95% of the variance.What does this tell you about the original 4-dimensional Iris dataset? From the findings, it is seen that most of the information contained and the variations found within all four variables are captured within just two dimensions. This shows the high inter-correlation found among the features, which makes it possible to reduce the dimensions without much information loss.